

Apollo AgriVerse: Technical Work Summary

Comprehensive Project Engineering Documentation for Industry Mentor Review

Project Domain: Precision Agronomy & Digital Twin	Date: August 2026
Target Crop Focus: Grapes (<i>Vitis vinifera</i>)	Target Platform: Apollo AgriVerse Engine
Technical Contact: Vijayalaxmi	Team Members: Sunaina, Dakshini, Nandini

1. Executive Project & Architecture Overview

Apollo AgriVerse is a multi-layered precision agriculture platform that integrates internet of things (IoT) telemetry, synthetic data generation, advanced ML predictive engines, deep-learning computer vision, dynamic frontend panels, and real-time backend API services to construct a dynamic **Digital Twin Engine** specifically tailored for grape farming (*Vitis vinifera*) in Indian agricultural conditions.

The project engineering workload was collaboratively executed across four core specialized modules: Synthetic Data Generation & Digital Twin Engine Architecture (**Vijayalaxmi**), Deep Learning Vision & Interactive Panels (**Sunaina Gaikwad**), High-Volume Spatial Weather Data Pipeline & Core Frontend System Panels (**Dakshini**), and Crop Dataset Phenology, NASA API, & FastAPI Microservices (**Nandini**). This document presents an explicit and detailed architectural record of all technical deliverables, file locations, mathematical methodologies, model parameters, and backend scripts.

2. Technical Work Summary — Vijayalaxmi

Role & Scope: Systems & Digital Twin Architect | Data Integration, ML Inference Engine, and Synthetic Framework

Vijayalaxmi architected the foundational data layer, synthetic data generation pipelines, master dataset relational synthesis, agronomic machine learning inference engines, and the digital twin telemetry simulation core powering Apollo AgriVerse.

2.1 Synthetic Datasets & Data Dictionary Specification

To model farm dynamics realistically, Vijayalaxmi authored the synthetic generator suite and data cleaning scripts, outputting five distinct raw/cleaned synthetic relational streams inside 02_Datasets/Synthetic/:

Dataset File	Key Schema Features	Value Ranges & Constraints
farm_metadata.csv (Raw & Clean)	farm_id, field_id, latitude, longitude, area_acres, soil_type, crop_type, installation_date	farm_id: F0001–F1000 Lat: 15.5–21.5, Long: 72.5–80.5 Area: 0.5–15 Acres Soil: Clay, Loamy, Black, etc. Crops: Cotton, Wheat, Rice, Grapes

sensor_stream.csv (Raw & Clean)	timestamp, farm_id, temperature, humidity, soil_moisture, soil_temperature, light_intensity, battery_level	Freq: Every 5 mins Air Temp: 18–42 °C Humidity: 30–95 % Soil Moisture: 15–90 % Soil Temp: 15–38 °C Light: 1,000–100,000 lux
hydrogel.csv (Raw & Clean)	timestamp, farm_id, water_storage, release_rate, remaining_capacity, status	Freq: 5 mins Water Storage: 0–100 % Release Rate: 0.2–5.0 L/hr Status: Healthy, Low, Critical
mulching.csv (Raw & Clean)	timestamp, farm_id, mulch_type, degradation_percent, evaporation_reduction, temperature_reduction, status	Mulch Types: Plastic, Organic, Biodegradable Degradation: 0–100 % Evap Reduction: 10–70 % Temp Reduction: 1–8 °C
crop_lifecycle.csv (Raw & Clean)	farm_id, crop_type, sowing_date, crop_age_days, growth_stage, plant_height_cm, leaf_area_index, chlorophyll_index, canopy_cover_percent	Age: 1–150 Days Height: 5–250 cm LAI: 0.5–6.0 Chlorophyll: 20–70 Canopy Cover: 5–100 %

2.2 Grape Focal Pivot & Master Integration Pipeline

Via Jupyter Notebook 02_Datasets/Master/build_grape_masters.ipynb, Vijayalaxmi synthesized two primary enterprise master datasets stored inside 02_Datasets/Processed/Grapes/:

- **Master 1: Master_Grapes_Agronomy_Reference.csv** (800 rows × 9 columns) — Integrates soil chemistry (NPK, pH) and microclimate requirements filtered specifically for grapes to power offline crop yield ML models.
- **Master 2: Master_Grapes_Digital_Twin_Telemetry.csv** (1,000 rows × 24 columns) — Aggregates 5-minute telemetry streams into daily state averages anchored on key farm_id, combining spatial metadata, growth stage phenology, hydrogel polymer metrics, and mulch film degradation.

2.3 Agronomic ML Machine Learning Engines

Vijayalaxmi engineered and trained two predictive machine learning models in 06_ML/notebooks/, serializing artifacts into 06_ML/models/ for live backend inference:

1. **Grape Crop Yield Prediction Model (grape_yield_model.pkl)**: Trained using Gradient Boosting Regressor / Random Forest Regressor on soil chemistry and weather parameters. Feature engineered parameters include N/K ratio, P/K ratio, and optimal pH deviation ($\text{ph_opt_dev} = |\text{soil_ph} - 6.5|$). Scaled via grape_yield_scaler.pkl.
2. **Telemetry Hydrogel Trigger Model (grape_telemetry_trigger_model.pkl)**: Predicts dynamic hydrogel water storage capacity requirement based on real-time sensor streams. Features engineered include $\text{thermal_gap} = \text{air_temp} - \text{soil_temp}$, $\text{effective_mulch_cooling}$, and $\text{evapotranspiration_index} = \text{air_temp} / (\text{humidity} + 1e-5)$. Scaled via grape_telemetry_scaler.pkl.

Sample ML Inference Verification Code (Engineered by Vijayalaxmi):

```
import joblib, pandas as pd, numpy as np
yield_model = joblib.load('../models/grape_yield_model.pkl')
yield_scaler = joblib.load('../models/grape_yield_scaler.pkl')
input_data = pd.DataFrame([{'nitrogen_mgkg': 140.0, 'phosphorus_mgkg': 45.0, 'potassium_mgkg': 210.0,
    'soil_ph': 6.4, 'air_temp_c': 26.5, 'humidity_pct': 65.0, 'rainfall_mm': 720.0,
    'N_K_ratio': 140.0/210.0, 'P_K_ratio': 45.0/210.0, 'ph_opt_dev': 0.1}])
scaled_input = yield_scaler.transform(input_data[yield_scaler.feature_names_in_])
pred_yield = yield_model.predict(scaled_input)[0] # Returns predicted yield (tons/ha)
```

2.4 Digital Twin Engine & Simulation Core

Vijayalaxmi developed the core simulation engine (located inside the project's simulation/ architecture) that models physical farm dynamic states in real time. It integrates real-time telemetry inputs, evaluates moisture loss against hydrogel polymer absorption curves, factors mulch film thermal cooling reductions, and triggers automated micro-irrigation interventions when soil moisture drops below critical physiological thresholds.

3. Technical Work Summary — Sunaina Gaikwad

Role & Scope: Deep Learning Vision Specialist & Frontend Interface Lead | Apollo AgriVerse Implementation

Sunaina Gaikwad focused on computer vision dataset curation, custom deep learning model training for soil, crop phenology, and polymer condition, alongside user dashboard frontend construction.

3.1 Tabular Data & Soil Dataset Engineering

Sunaina developed dedicated Python scripts to generate localized tabular soil datasets mimicking Indian regional soil and telemetry profile conditions. All synthetic soil practice files were stored directly inside 02_Datasets/.

3.2 Image Dataset Curation & Folder Structuring

- **Regional Indian Soil Image Dataset:** Sourced from Kaggle and structured under 02_Datasets/Processed/Soil/Image_Dataset/ into train, val, and test splits covering 7 major classes: *Alluvial_Soil*, *Arid_Soil*, *Black_Soil*, *Laterite_Soil*, *Mountain_Soil*, *Red_Soil*, and *Yellow_Soil*.
- **Hydrogel & Mulch Condition Image Dataset:** Generated and preprocessed synthetic/real image datasets stored under 03_Images/Dataset/ to inspect state transitions in water-retention polymers (dry vs. swollen hydrogel) and surface mulching film integrity (intact vs. degraded film).

3.3 Deep Learning AI Models Built & Trained (.keras)

Model Name	Architecture & Methodology	Notebook & Model Paths	Functional Purpose
Soil Classification Model	MobileNetV2 Transfer Learning fine-tuned for 7 classes	Notebook: 06_ML/notebooks/train_soil_model.ipynb Model: 06_ML/models/soil_classification_model.keras	Classifies input soil images into one of 7 regional Indian soil types to inform agronomic baseline setup.
Grape Crop Stage Model	Convolutional Neural Network (CNN) / Transfer Learning	Notebook: 06_ML/notebooks/train_crop_model.ipynb Model: 06_ML/models/grape_crop_stage_model.keras	Identifies grapevine growth across 5 life stages (Bud Burst, Vegetative, Flowering, Fruit Dev, Harvest).
Hydrogel & Mulch Inspection Model	Custom CNN for state integrity & surface classification	Notebook: 06_ML/notebooks/train_hydrogel_model.ipynb Model: 06_ML/models/hydrogel_model.keras	Inspects physical state of polymers (dry/swollen) and mulch film degradation (intact/damaged).

3.4 Frontend Dashboard Panel Development

Utilizing UI generation platforms (v0, ChatGPT, Gemini, Codex), Sunaina designed and implemented **2 out of the 6 core platform panels** for the AgriVerse user dashboard:

- 1. **Smart Mulching Panel:** Built to monitor real-time physical condition, degradation level, coverage percentage, and moisture retention efficiency of protective ground films.
- 2. **Predictions Panel:** Constructed to display live AI computer vision classifications, growth stage progression, yield predictions, and operational system alerts directly to end-users.

4. Technical Work Summary — Dakshini

Role & Scope: High-Volume Spatial Weather Data Pipeline Architect & Core Frontend System Lead

Dakshini engineered the large-scale weather data processing pipeline (managing over **800,000 continuous entries**) and spearheaded the frontend design and implementation of the **6 Main Core Platform Panels** that form the primary interactive backbone of Apollo AgriVerse.

4.1 Core System Frontend Panels Architecture (6 Main Panels)

Dakshini led the UI/UX architecture, component structure, and state integration for the 6 central system panels across the platform:

Core Panel Name	Key Functional & Interface Components	Data Integration & Outputs
1. Digital Twin Panel	Interactive 3D/2D spatial farm visualizer, telemetry node indicators, live sensor metric gauges.	Streams real-time sensor streams (temp, moisture, light) into a live virtual farm model.
2. Crop Life Cycle Panel	Phenology progress indicators, growth timeline (Bud Burst to Harvest), leaf area index & height graphs.	Binds with crop_lifecycle.csv and vision model stage progression data.
3. Simulation Panel	Interactive environment controls (irrigation rates, temp offset, fertilizer inputs), scenario execution triggers.	Interfaces directly with the Digital Twin dynamic physics simulation engine.
4. Hydrogel & Smart Irrigation Panel	Polymer hydration levels, release rate meters, automated micro-irrigation valve toggles.	Monitors hydrogel water storage and triggers automated irrigation interventions.
5. Weather Analytics Panel	Spatial microclimate maps, hourly temperature/humidity trends, precipitation forecasting charts.	Powered by Dakshini's 800k Open-Meteo pipeline and live NASA weather API streams.
6. Executive Dashboard Panel	High-level KPI metric cards, master system health monitoring, critical alert feed.	Aggregates real-time state streams across all farm sub-systems into a unified view.

4.2 High-Volume Weather Data Pipeline (800,000+ Entries)

Environment & Python Tech Stack:

- ``requests`` & ``urllib3``: Developed robust API client connection logic with automated retries and network fault tolerance.
- ``pandas`` & ``numpy``: Executed high-performance vector math, time-series indexing, and structured data manipulation on 800k rows.

- ``scikit-learn`` (``RobustScaler``): Applied robust feature scaling based on statistics that are resilient to outliers, preserving critical extreme weather event signals (e.g., severe heatwaves, sudden rain spikes).
- ``time`` & ``os``: Managed rate-limiting algorithms to respect external API throttle policies and orchestrated local system file saving.

External Data Sourcing & Advanced ML Techniques:

- **Open-Meteo Archive API:** Ingested 2023 historical hourly weather telemetry across a customized latitude/longitude grid covering key Indian agricultural zones.
- **Time-Weighted Interpolation:** Applied linear and spline temporal interpolation algorithms to resolve missing continuous data points without destroying temporal continuity.
- **Outlier Detection & Noise Smoothing (IQR & EWMA):** Utilized Interquartile Range (IQR) filtering to isolate physical sensor anomalies and Exponentially Weighted Moving Average (EWMA) to smooth high-frequency microclimate sensor noise.
- **Cyclical Feature Encoding:** Transformed temporal features (hours of day, months of year) into continuous 2D spatial features using Sine and Cosine trigonometric transformations:

```
x_sin = sin(2 * pi * t / T), x_cos = cos(2 * pi * t / T)
```

- **Time-Series Windowing & Feature Extraction:** Engineered 1-hour, 3-hour, and 24-hour lag variables, alongside 6-hour rolling means and rolling standard deviations to provide rich temporal context for downstream ML models.

5. Technical Work Summary — Nandini

Role & Scope: Crop Phenology Pipeline, NASA API Testing, & FastAPI Microservice Engineer

Nandini delivered work spanning crop dataset organization, grape phenology image preprocessing, growth stage model training, external NASA POWER API integration, and setting up the central FastAPI backend server.

5.1 Crop Dataset & Image Phenology Pipeline

- **Agronomic Crop Recommendation Dataset:** Handled and verified `02_Datasets/Raw/Crop/Crop_recommendation.csv` for multi-crop comparison baselines.
- **Grape Crop Image Structuring:** Curated and organized grape plant images under `03_Images/Dataset/` into 5 distinct biological growth directories: `01_Bud_Burst`, `02_Vegetative`, `03_Flowering`, `04_Fruit_Development`, and `05_Harvest`.

5.2 Computer Vision Preprocessing & Model Execution

- Preprocessed image datasets using OpenCV, NumPy, and TensorFlow/Keras to verify image resolutions, channels, and bounding integrity.
- Built and trained the **Grape Crop Stage Classification Model** inside notebook `06_ML/notebooks/train_crop_model.ipynb`, saving the final model artifact to `06_ML/models/grape_crop_stage_model.keras`.

5.3 External Weather API Integration & Verification

- Tested real-time solar and meteorological data extraction using the **NASA POWER Hourly API** across target Indian coordinates.

- Developed API client logic in 05_Backend/api/weather_api/nasa_api.py and automated verification test script in 05_Backend/api/weather_api/test_nasa_weather.py, extracting temperature, relative humidity, and precipitation parameters.

5.4 FastAPI Microservice Architecture

- Constructed and tested the core FastAPI application entry point inside 05_Backend/app/main.py.
- Configured Uvicorn ASGI server deployment and verified backend endpoint routing via Interactive OpenAPI / Swagger documentation.
- Maintained team version control across Git/GitHub repositories.

Nandini's End-to-End Workflow Vector:

Grape Dataset → Image Preprocessing (OpenCV) → Grape Growth Stage Model (.keras) → NASA Weather API Engine → FastAPI Service Setup

6. Team Task & Technical Ownership Matrix

The following table provides a high-level summary of technical deliverables and individual task ownership across Apollo AgriVerse:

Project Subsystem	Primary Engineer	Key Deliverables & Technology Stack
Synthetic Data & Digital Twin	Vijayalaxmi	5 Synthetic CSVs, Master Agronomy & Telemetry Pipelines, Digital Twin Simulation Core, Yield & Hydrogel ML Engines (`joblib`, `scikit-learn`).
6 Main Core Frontend Panels & Weather Pipeline	Dakshini	Designed & built 6 main panels (Digital Twin, Life Cycle, Simulation, Hydrogel, Weather, Overview Dashboard). 800,000-entry Open-Meteo processing (`RobustScaler`, EWMA/IQR).
Soil & Polymer Vision Models	Sunaina Gaikwad	Soil & Mulch Image Datasets, 3 Keras DL Models (MobileNetV2, CNNs), Smart Mulching & Predictions UI Panels.
Crop Phenology & Backend API	Nandini	Grape Phenology Dataset, Crop Stage Classification Model, NASA POWER API client, FastAPI (`main.py`) & Uvicorn setup.

7. Document Authorization & Sign-off

Submitted By: Vijayalaxmi Systems & Digital Twin Architect	Reviewed By: Industry Mentor Apollo AgriVerse Evaluation
Signature: _____	Signature: _____